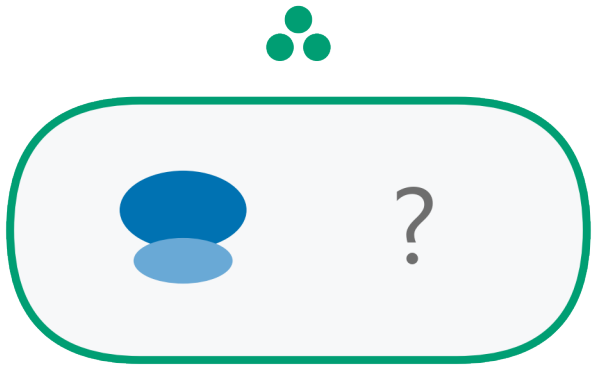
Direction of change in the ribosomal fraction of E. coli under nutrient limitation and translation inhibition

CCBS 2026 · Lecture 14 · The growth biomachine

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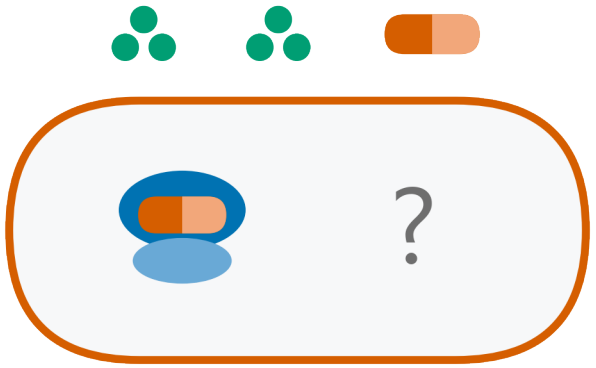
Poorer nutrients and chloramphenicol both slow E. coli growth — how does each change the ribosomal fraction?



M63 + glycerol
no drug
nutrient-poor medium
without any drug

$\lambda = 0.40 \pm 0.03\ h^{-1}$

≈



cAA + glucose
+ 8 μM chloramphenicol
richer medium with
chloramphenicol

$\lambda = 0.43 \pm 0.06\ h^{-1}$

Let λ = growth rate (h^{-1}), doubling time

$$t_d = \frac{\ln 2}{\lambda} \qquad \lambda = 0.40 \rightarrow t_d \approx 104 \text{ min}$$

Same λ within error: One is limited by nutrients, while the other is limited by translation.

Ribosome: Ribosomes are responsible for protein synthesis

Chloramphenicol: an antibiotic; binds ribosomes and slows down translation

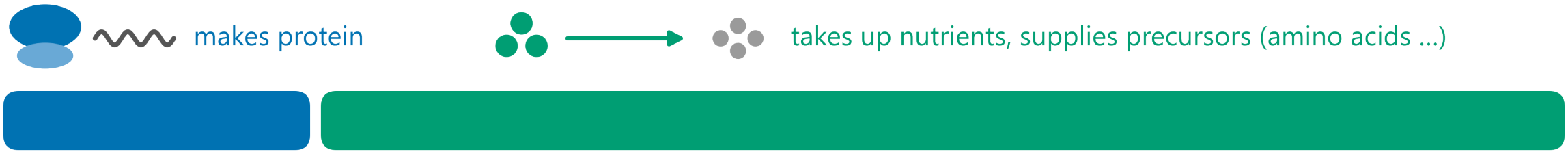
Ribosomal fraction: share of cellular protein in ribosomes and translation-related proteins.

Question: *does ϕ_R change in the same way in these two cultures?*

Point: *both lower λ but move ϕ_R in opposite directions; the supply–translation balance can explain why.*

Data: Scott et al., Science 330:1099 (2010), SOM Tables S1, S3, S4 (strain EQ2)

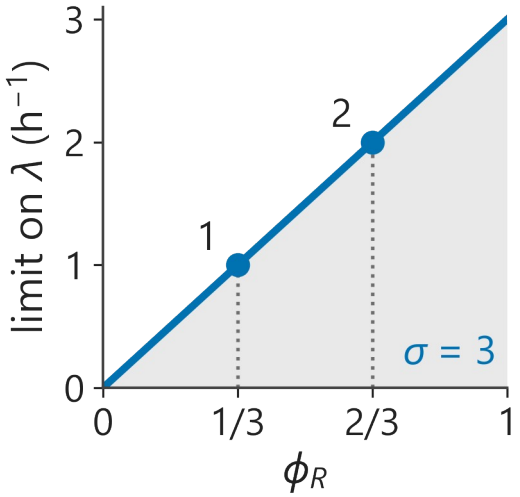
Two sectors of cell protein, two limits on growth



ribosomal sector ϕ_R

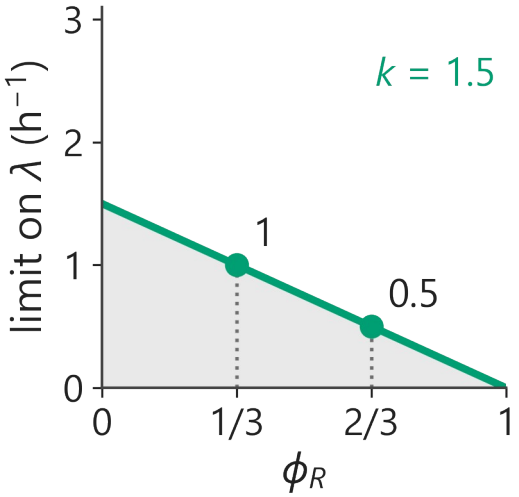
$$\phi_R = \frac{M_R}{M}, \qquad \frac{M_P}{M} = 1 - \phi_R$$

metabolic sector $1 - \phi_R$



Let σ = translation capacity: the protein made per gram ribosomal sector per hour

$$\lambda = \frac{\dot{M}}{M} \leq \sigma\phi_R$$



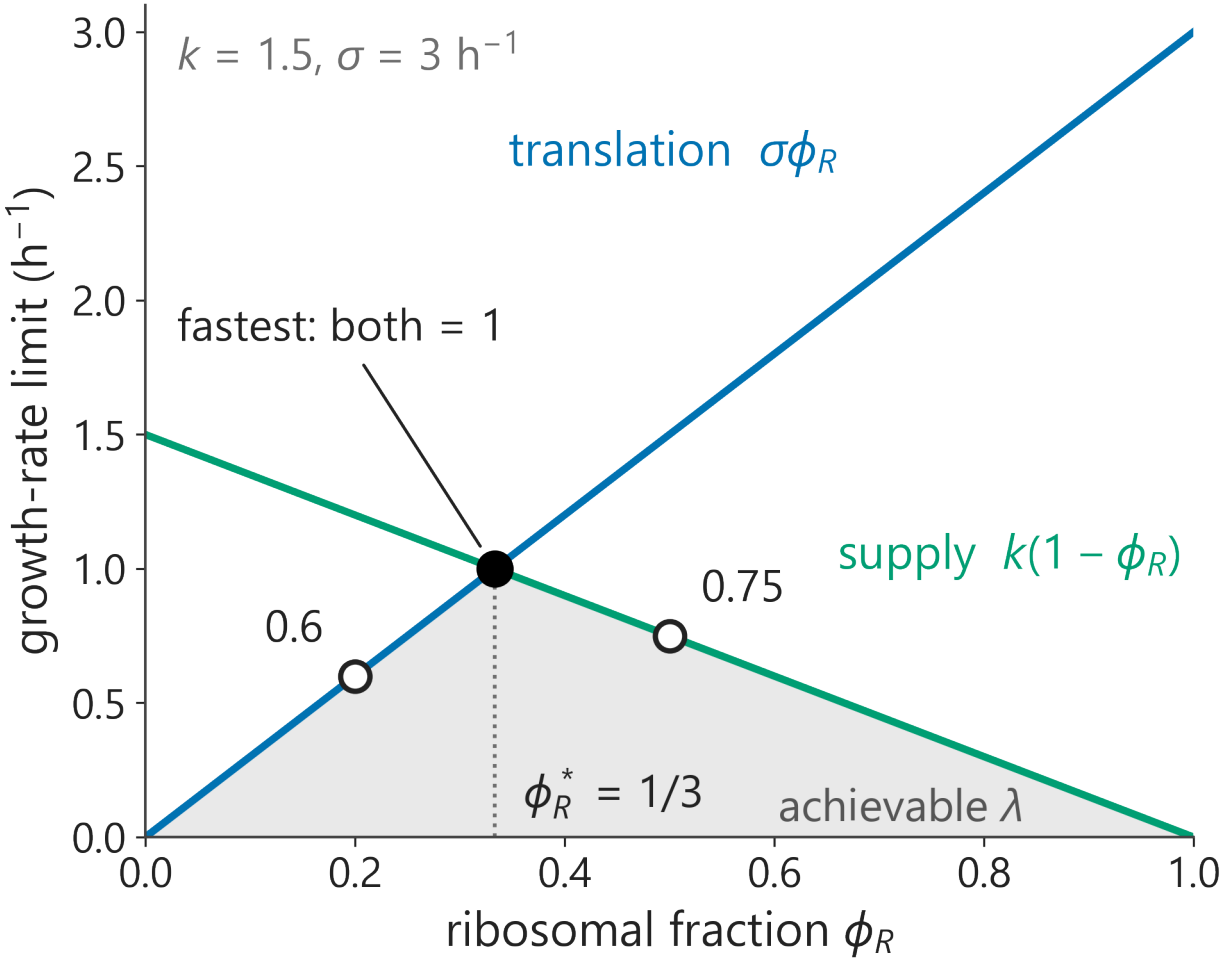
Let k = supply capacity: the precursor supplied per gram metabolic sector per hour

$$\lambda \leq k(1 - \phi_R)$$

Chloramphenicol lowers *σ, and* better nutrients raise *k*.

Illustration: two-sector model; numbers are illustrative

The two sectors share one protein budget; growth is fastest when the two limits are equal



Since both limits hold:

$$\lambda = \min\{\sigma\phi_R,\ k(1-\phi_R)\}$$

ϕ_R = 0.2 → 0.6 (open dot)

ϕ_R = 0.5 → 0.75 (open dot)

Fastest ⇔ the two limits are equal:

$$\sigma\phi_R^* = k(1-\phi_R^*)$$

$$\phi_R^* = \frac{k}{k+\sigma}, \qquad \lambda^* = \frac{k\sigma}{k+\sigma}$$

k = 1.5, σ = 3 ⇒

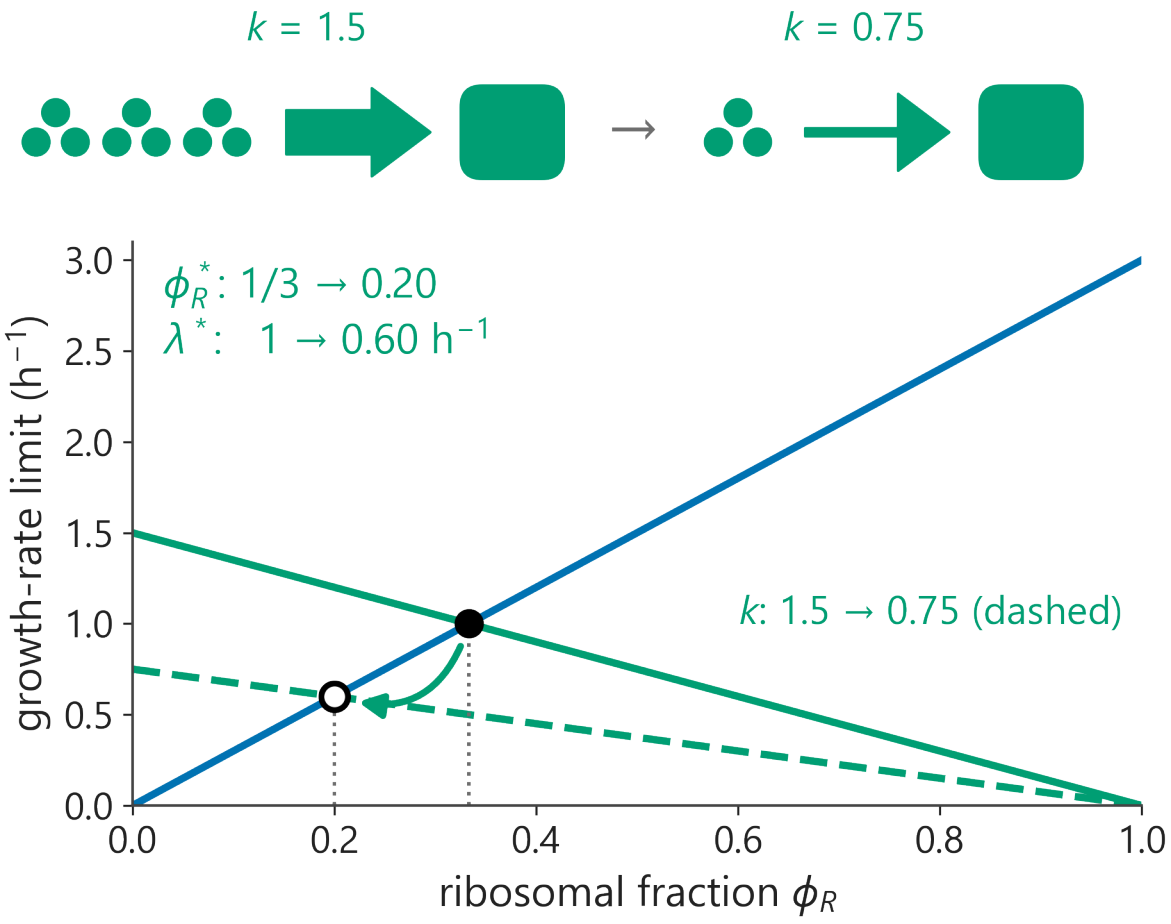
ϕ_R^* = 1/3, λ^* = 1 h^{-1} (black dot)

Assume: the cell sits at this optimum

→ The rest follows from how it moves

Illustration: two-sector model; numbers are illustrative

Poorer nutrients weaken supply capacity: the ribosomal fraction falls with growth



Poorer nutrients ⇒ Each gram of metabolic sector supplies fewer precursors ⇒ k ↓; **assume** σ fixed

$$\phi_R^* = \frac{1}{1+\sigma/k}, \qquad \lambda^* = \frac{1}{1/k+1/\sigma}$$

k ↓ ⇒ σ/k ↑ ⇒ ϕ_R^* ↓

k ↓ ⇒ $1/k$ ↑ ⇒ λ^* ↓

k: 1.5 → 0.75 ⇒ ϕ_R^*: 1/3 → 0.20, λ^*: 1 → 0.60

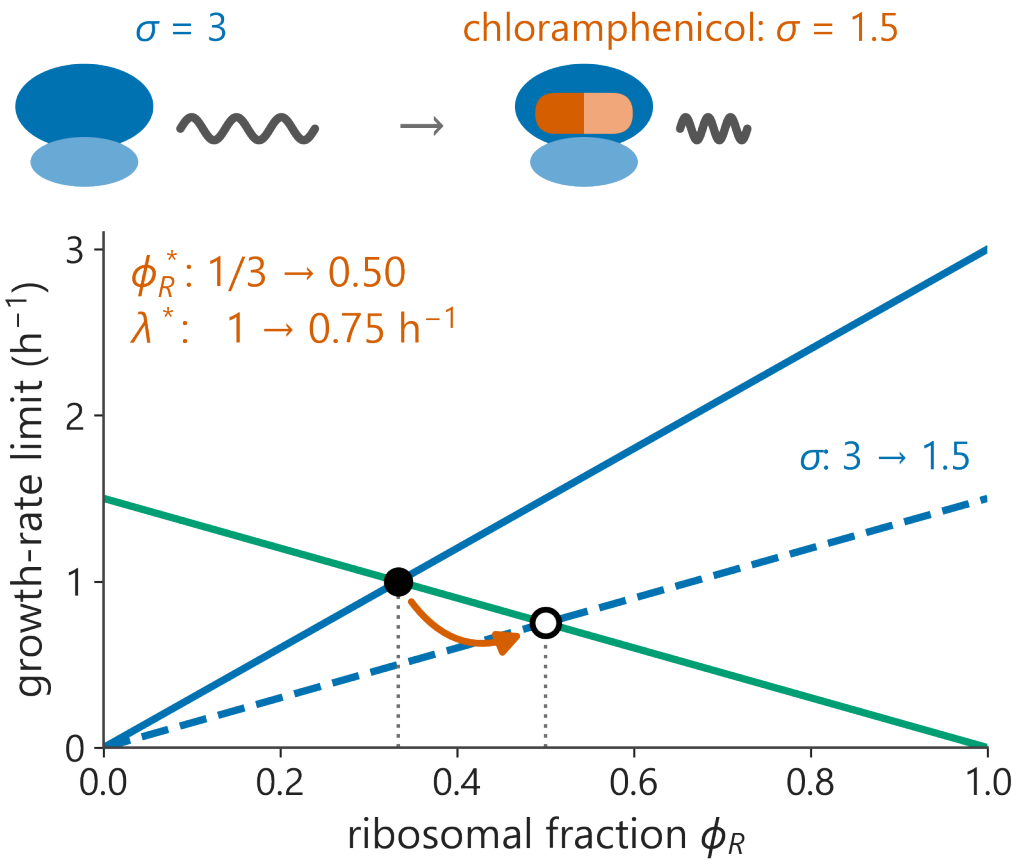
σ fixed ⇒ optimum lies on the translation limit $\lambda^* = \sigma\phi_R^*$:

$$\phi_R^* = \frac{\lambda^*}{\sigma}$$ line through 0

So with poorer nutrients, ϕ_R falls with growth

Illustration: two-sector model; numbers are illustrative; σ = 3 h^{-1} fixed

Chloramphenicol weakens translation capacity: the ribosomal fraction rises instead



Chloramphenicol binds the ribosome and slows translation ⇒ so σ falls ↓, and we assume *k stays* fixed (same medium)

$$\phi_R^* = \frac{1}{1+\sigma/k}, \qquad \lambda^* = \frac{1}{1/k+1/\sigma}$$

$\sigma \downarrow \Rightarrow \sigma/k \downarrow \Rightarrow \phi_R^* \uparrow$

$\sigma \downarrow \Rightarrow 1/\sigma \uparrow \Rightarrow \lambda^* \downarrow$

σ: 3 → 1.5 ⇒ ϕ_R^*: 1/3 → 0.50, λ^*: 1 → 0.75

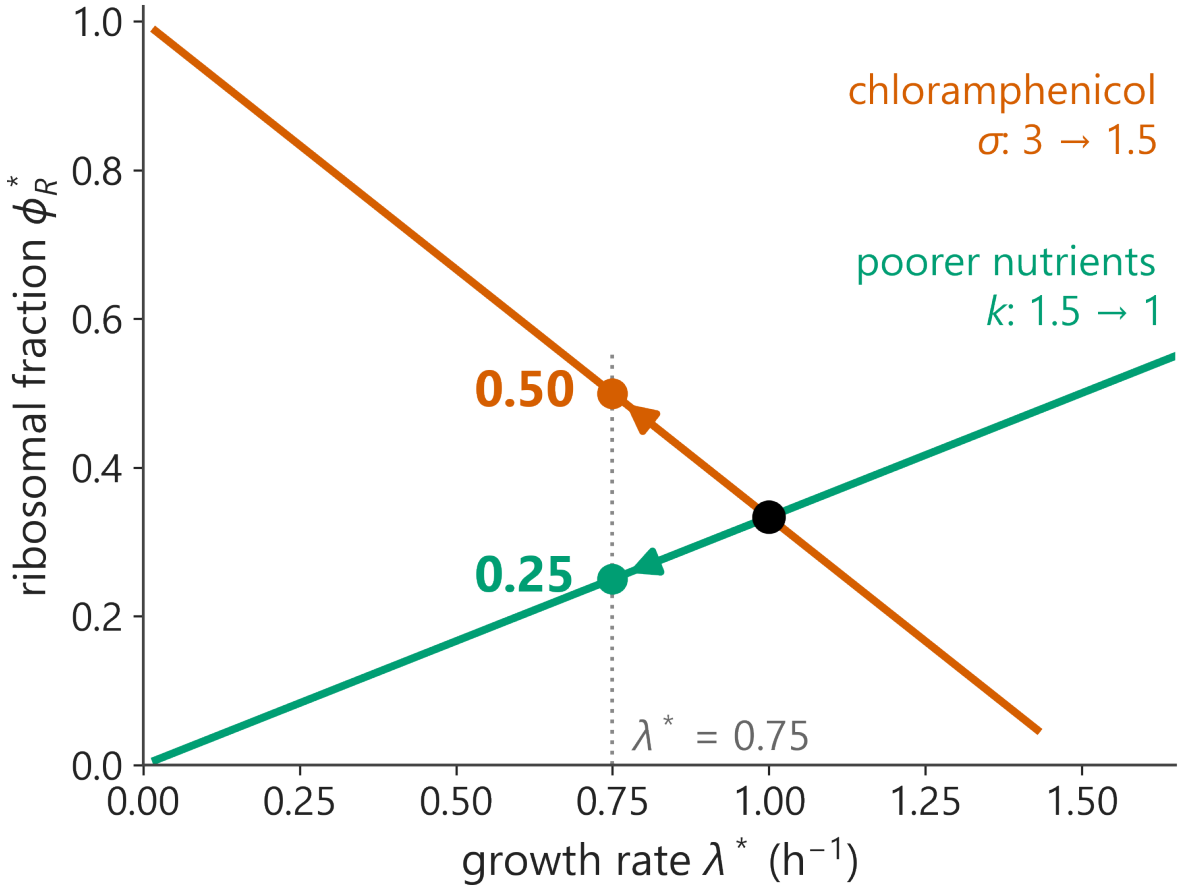
k fixed ⇒ every optimum lies on the supply limit $\lambda^* = k(1 - \phi_R^*)$:

$$\phi_R^* = 1 - \frac{\lambda^*}{k}$$ falling; $\phi_R^* = 1$ at $\lambda^* = 0$

So chloramphenicol raises ϕ_R, while growth still falls

Illustration: two-sector model; numbers are illustrative; k = 1.5 h^{-1} fixed

Hitting k or σ gives the same drop in growth but opposite changes in the ribosomal fraction



Start: black dot, k = 1.5, σ = 3

Slow both routes to λ^* = 0.75:

k: 1.5 → 1: ϕ_R^* = 0.75/3 = **0.25**

σ: 3 → 1.5: ϕ_R^* = 1 − 0.75/1.5 = **0.50**

Prediction

poorer nutrients reduce growth and ϕ_R

chloramphenicol also slows growth but increases ϕ_R

⇒ At the same growth rate, the ribosomal fraction is twice as high with chloramphenicol.

→ test with RNA/protein data

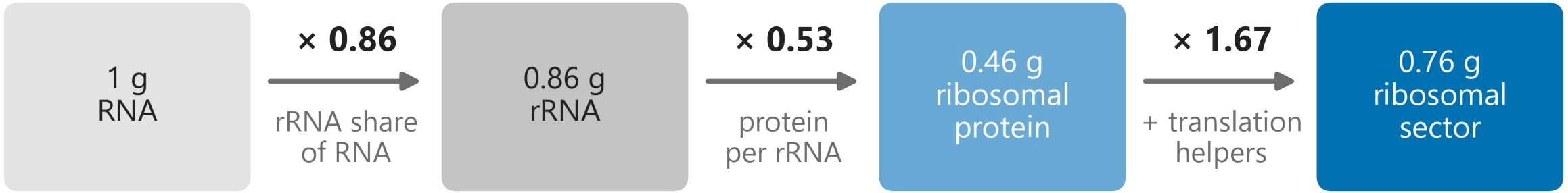
Illustration: two-sector model; numbers are illustrative; black dot: k = 1.5, σ = 3 h^{-1}

Experiments use the RNA/protein ratio to measure the ribosomal fraction

To test this prediction, we need ϕ_R, but it is hard to measure directly, so we can measure the RNA to protein ratio, r.

Most RNA in the cell is ribosomal RNA, so the more RNA there is, the more ribosomes there are.

$r = \frac{M_{\mathrm{RNA}}}{M}$ **Let** *r* = RNA/protein $\quad \phi_R \approx \rho\, r$ e.g. $r = 0.287 \rightarrow \phi_R \approx 0.22$ $\quad \rho = 0.86 \times 0.53 \times 1.67 \approx 0.76$

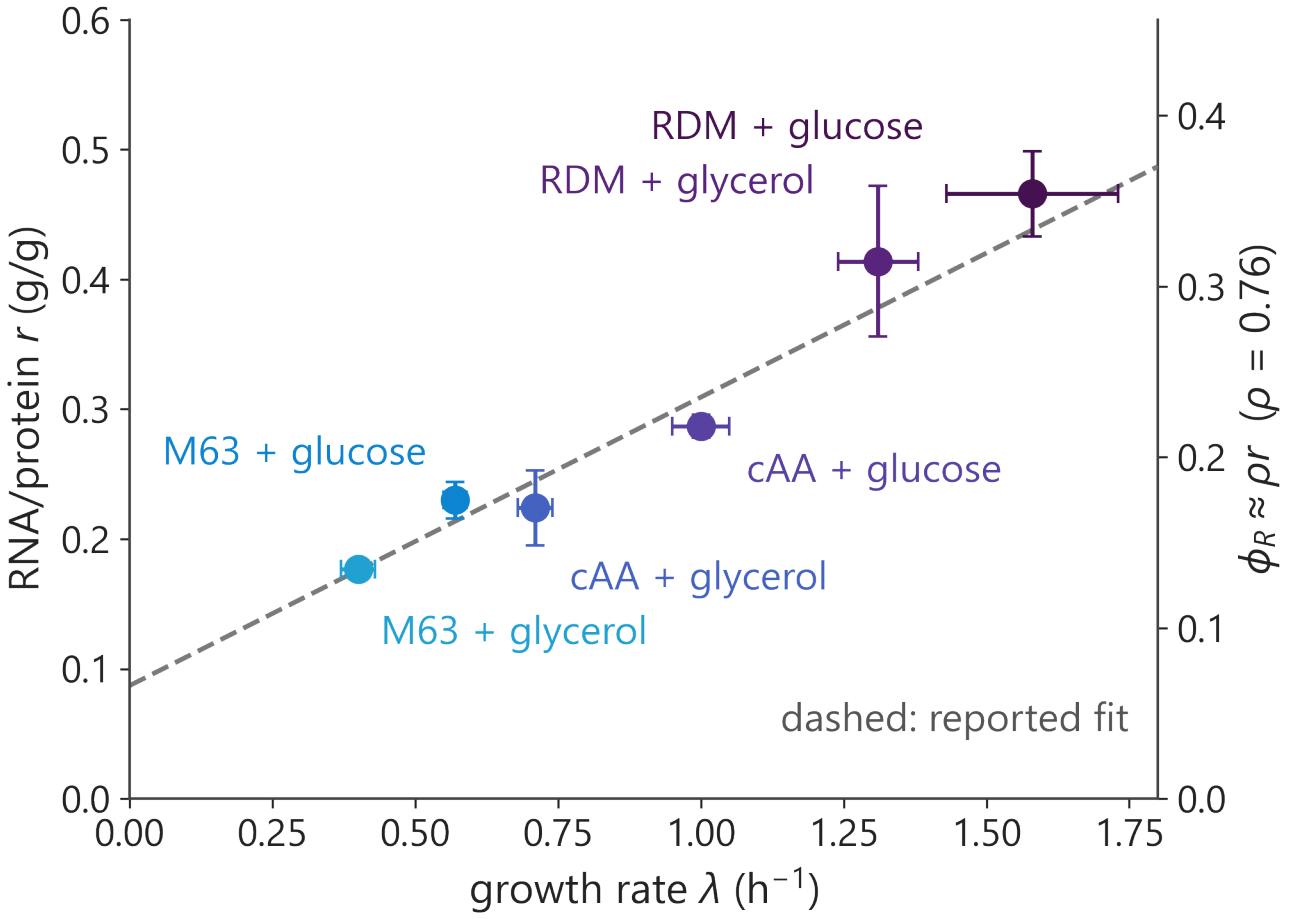


$\mathrm{fold}(\phi_R) = \mathrm{fold}(\rho) \times \mathrm{fold}(r)$ If this factor satys constant , ϕ_R moves the same way as r.

So in the data, we read the direction of ϕ_R from *r*.

Conversion factors: Scott et al. (2010) SOM Eq. S1; fold changes: SOM Table S3

Changing only the medium: the RNA/protein ratio rises with growth



Only change the medium ⇒ mainly changes k, **while** $\sigma \approx$ fixed

⇒ So the model predicts r rises with λ

EQ2, 37 °C, ≥ 3 generations of steady growth;
Across six media, blue (poor) → purple (rich)

λ: 0.40 → 1.58 h^{-1}; r: 0.177 → 0.466

$$r = r_0 + \frac{\lambda}{\quad}$$

This trend is consistent with the model's prediction.

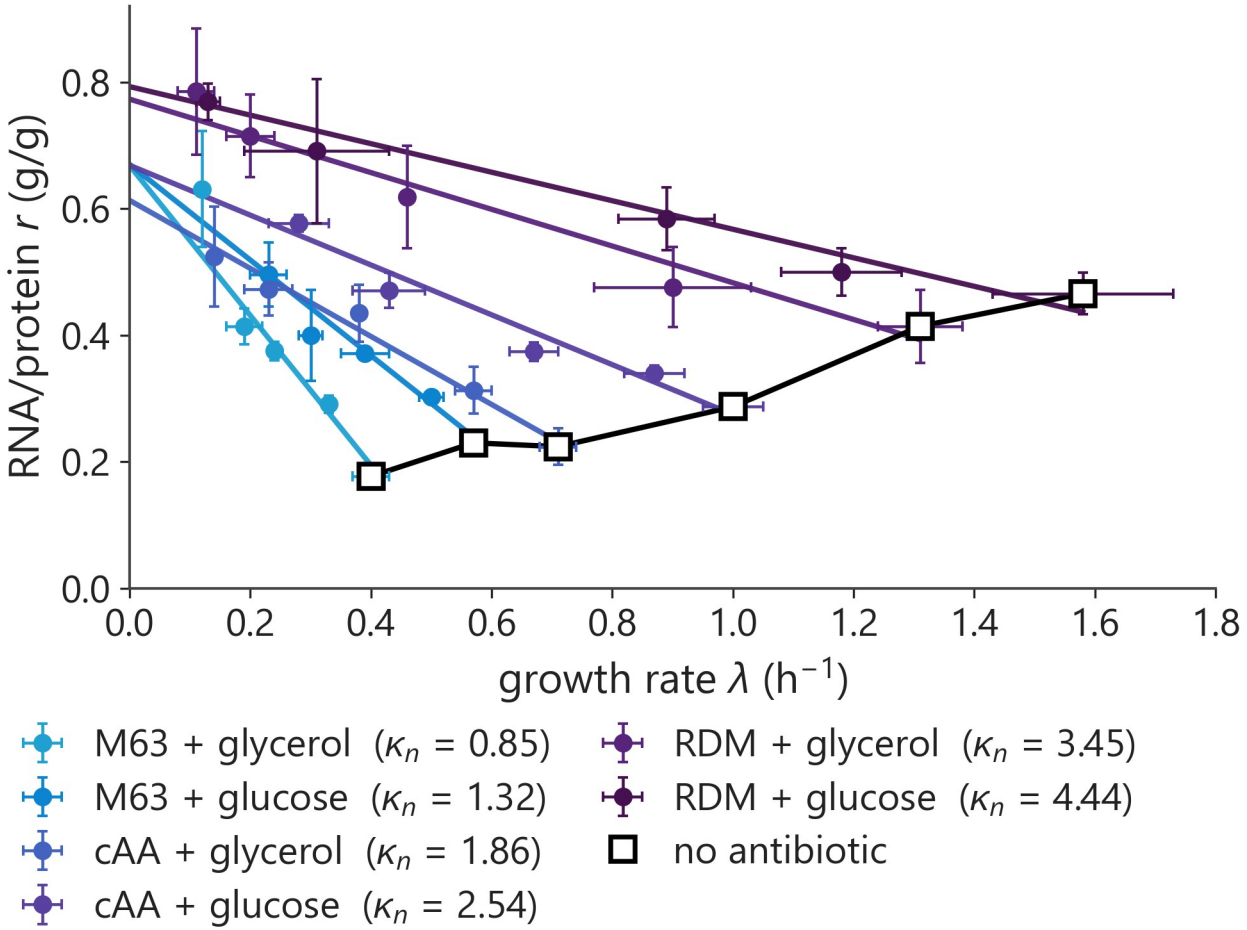
right axis: $\phi_R \approx 0.76\, r$

Hence same direction as predicted

But $r_0 \neq 0$ (model line through 0)

Data: Scott et al., Science 330:1099 (2010), SOM Tables S1, S3, S4; dashed: fit reported in SOM Table S1

Adding chloramphenicol: growth slows, but the RNA/protein ratio rises



Add chloramphenicol to the same medium

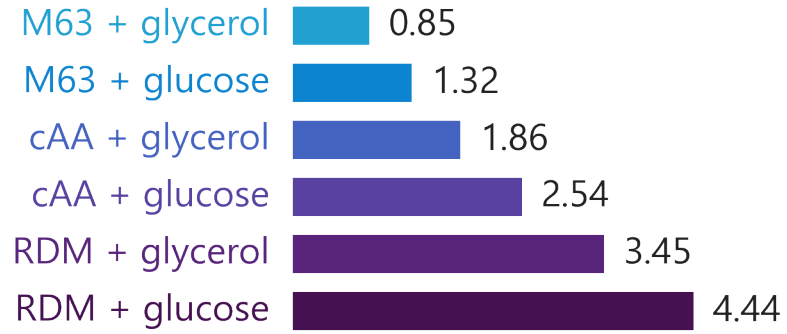
⇒ which lowers σ while k stays fixed

⇒ So the model predicts r rises ↑ as λ *falls* ↓

All six media: more chloramphenicol ↑ ⇒ λ ↓, r ↑

$$r = r_{\max} - \frac{\lambda}{\kappa_n}$$ per medium; κ_n = 1/|slope|

Model slope $-1/k$ ⇒ richer medium, flatter line:

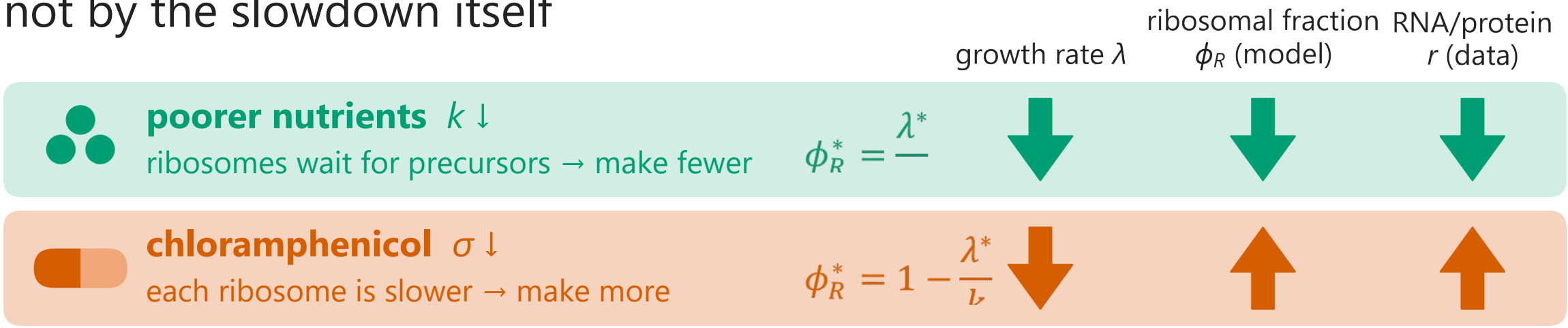


Two cultures (slide 2): r = 0.177 vs **0.471** (2.66×)

Data: Scott et al., Science 330:1099 (2010), SOM Tables S1, S3, S4; lines: fits counting errors in both λ and r

So the supply–translation optimum explains why the two perturbations move the ribosomal fraction in opposite directions

The direction of ϕ_R is set by which capacity is hit (k or σ),
not by the slowdown itself



The supply–translation optimum predicts both directions, and the data agree.

M63 + glycerol, no drug

λ = 0.40 h^{-1}, r = 0.177 $\phi_R \approx$ **0.13**

same $\lambda \approx$ 0.4 h^{-1}

ϕ_R 0.13 vs 0.36 (2.7×)

cAA + glucose + 8 µM chloramphenicol

λ = 0.43 h^{-1}, r = 0.471 $\phi_R \approx$ **0.36**

Model + data: Scott et al. (2010) SOM Table S3; $\phi_R \approx 0.76\, r$